



COURSE DESCRIPTION		
Course Code	26SE102	Course Credit: 3
Course Name	Advanced Data Structures & Algorithm Design	
Course Delivery Mode	TiC/MOOC	
L-T-P-H	3-0-0-0	
Pre - requisite	Nil	
Year of Introduction and Syllabus Version	2026 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	To introduce fundamental concepts of algorithm design, complexity analysis, and asymptotic behaviour.	L2	K1
2	To develop understanding of core data structures such as stacks, queues, trees, and their applications.	L3	K2
3	To enable the application of advanced data structures and algorithm design paradigms like Divide and Conquer and Greedy methods.	L3	K3
4	To analyse and evaluate graph algorithms, optimization techniques, and advanced sorting and selection methods.	L4	K4

COURSE OUTCOMES (COs)	
After the successful completion of the course students will be able to:	
CO1	Explain algorithmic concepts, complexity measures, and asymptotic analysis techniques.
CO2	Apply fundamental data structures such as stacks, queues, and binary search trees to solve problems.
CO3	Design algorithms using advanced paradigms like Divide and Conquer and implement balanced trees and heaps.
CO4	Analyse graph algorithms and advanced techniques for solving optimization and real - world computational problems.

CO - PO MAPPING	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	2	-	-	1
CO2	3	1	3	-	1	2
CO3	3	2	3	-	2	2
CO4	3	2	3	1	2	2

Correlation levels: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"



COURSE IMPLEMENTATION CLOCK (IN HOURS)	
Zeroth Module	1 Hr.
Course content delivery hours	36 Hrs.
Internal Test (2 Assessment Tests of one and a half hour duration)	3 Hrs.

COURSE CONTENTS

Zeroth Module: General Introduction to the Course		
- Course introduction with a thorough briefing of Course Objectives and Course Outcomes - An overview of the POs that are mapped to the course	No ESE	1 Hr.

Module - I		COs	Hrs.
Module Title	Introduction to Algorithm Design and Analysis		
Introduction to algorithmic thinking and characteristics of efficient algorithms - Time and Space Complexity analysis - Asymptotic notations - Linear - time algorithms - Maximum subarray problem - Mathematical foundations of algorithm analysis - Techniques for writing and proving correctness of algorithms - Local minima problems in 1D and 2D arrays.		1	9
Module - II		COs	Hrs.
Module Title	Fundamental Data Structures		
Abstract data types (ADT) - Range minimum query (RMQ) in 1D arrays - Binary search trees (BST) - Structure and properties - Height and balancing concepts - Stack and queue - Expression evaluation using stacks - Grid - based distance problems		2	9
Module - III		COs	Hrs.
Module Title	Advanced Data Structures and Algorithmic Paradigms		
Balanced trees - Red - black trees - Height - Insertion - Deletion operations - Divide and conquer paradigm - Applications - Half - majority problem - Efficient multiplication techniques - Inversion counting - Heap data structure and priority queues - Dynamic data structures		3	9
Module - IV		COs	Hrs.
Module Title	Graph Algorithms and Advanced Techniques		
Graph representations and fundamentals - Traversal algorithms - Breadth first search (BFS) - Depth first search (DFS) and DFS tree - Bipartite graphs - Articulation points and biconnected components - Disjoint sets (Union - Find) - Set operations - Dijkstra's shortest path algorithm - Selection problems - Non - comparison sorting - Range queries and multi - increment techniques.		4	9

TEXT BOOKS	
1	T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, <i>Introduction to Algorithms</i> , 4 th edition Cambridge, MIT Press, 2022, ISBN: 978 - 0262046305,
2	M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, <i>Data Structures and Algorithms in Java</i> , 6 th edition Hoboken, NJ, USA: Wiley, 2014, ISBN: 978 - 1118771334,



3	J. Kleinberg and Eva Tardos, <i>Algorithm Design</i> , 2 nd edition Harlow, U.K.: Pearson Education, 2022, ISBN: 978 - 0132131087,
REFERENCE BOOKS/ WEB REFERENCES	
1	R. Sedgewick and K. Wayne, <i>Algorithms</i> , 4 th edition Boston, MA, USA: Addison - Wesley, 2011, ISBN: 978 - 0321573513,
2	S. Dasgupta, C. Papadimitriou, and U. Vazirani, <i>Algorithms</i> , 1 st edition, New York, NY, USA: McGraw - Hill, 2006, ISBN: 978 - 0073523408,
3	A. V. Aho, J. E. Hopcroft, and J. D. Ullman, <i>Data Structures and Algorithms</i> , 1 st edition, Reading, MA, USA: Pearson, 1983, ISBN: 978 - 0201000238,
4	S. S. Skiena, <i>The Algorithm Design Manual</i> , 3 rd edition Cham, Switzerland: Springer, 2020, ISBN: 978 - 3030542559,
5	S. Baase and A. Van Gelder, <i>Computer Algorithms: Introduction to Design and Analysis</i> , 3 rd edition Boston, MA, USA: Pearson, 2000, ISBN: 978 - 0201612448,

MODES OF EVALUATION	SCORE WEIGHTAGE / SPLIT MARKS
Continuous Internal Evaluation (CIE)	40
Internal Test - I	12.5
Internal Test - II	12.5
Task - works	15
Assignment	50%
Seminar	50%
End Semester Examination (ESE)	60
Total	100
End Semester Examination Pattern	
The question paper consists of two parts: Part A: 24 marks Part B: 36 marks Part A contains a total of 8 questions, 2 question each from each module, having 3 marks. Part B contains a total of 8 questions, 2 questions from each module of which students should answer any one. Each question in part B carries 9 marks and can have a maximum of 2 sub - divisions.	

APPROVAL SUMMARY	
	Number and Date
Recommended by BoS	SCE/BoS/CS/M.Tech./SE/013/25- 26, 22-05-26
Approved by Academic Council	